Image Plagiarism Detection with Equivariant Encoders

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Abstract

Image plagiarism detection should remain robust to rotations, reflections, and integer shifts of the pixel grid. We ask whether encoding these symmetries as an architectural constraint outperforms approximating them with data augmentation. We train a siamese detector with equivariant ViT encoders under three symmetry groups - D_4, $\mathbb{Z}^2$, and $SE(2)$ - combined with an L^2 contrastive regularizer, and compare them against an augmented ViT-L/16 baseline on a balanced DomainNet protocol. We report per-transformation FPR and Recall across 10 training runs covering five encoders and two regularization conditions.

1 Introduction

Image plagiarism detection asks whether two images x and x' depict the same underlying content after one of them has been transformed geometrically or photometrically. The question we study is direct: does building geometric symmetry into the encoder architecture improve plagiarism detection over relying on data augmentation alone? We compare five encoder configurations across three symmetry groups, D_4, $\mathbb{Z}^2$, and $SE(2)$, within a single siamese framework, training each with and without an L^2 contrastive regularizer on a balanced cross-domain benchmark.

The objects of study are pairs of natural images related by transformations drawn from D_4 (rotations by multiples of 90° and reflections), $\mathbb{Z}^2$ (integer pixel shifts), or $SE(2)$ (arbitrary rotations and translations), compounded with photometric perturbations such as brightness jitter, contrast scaling, and JPEG compression. Standard Vision Transformers (ViTs) [1] lack geometric invariance by construction: patch-embedding tokenization is not equivariant to rotations or reflections, and self-attention treats spatial position through learned positional encodings that do not extrapolate well outside the training distribution. A common workaround is geometric augmentation, but augmentation only stochastically covers the orbit of a transformation group, inflates training cost, and offers no invariance guarantee on unseen perturbations. Model scale does not resolve the underlying limitation: recent evidence indicates that spatial and geometric reasoning fails to improve with scale alone [2]. This motivates embedding the desired symmetry as an inductive bias in the architecture itself, along the lines of geometric deep learning [3, 4].

Formally, an encoder $\mathrm{phi}_{\mathrm{theta}} : \mathcal{X} \to \mathcal{Z}$ is *G-equivariant* if $\mathrm{phi}_{\mathrm{theta}}(g \cdot x) = \mathrm{rho}(g)\,\mathrm{phi}_{\mathrm{theta}}(x)$ for all $g \in G$ and all images x, where rho is a representation of G on the embedding space $\mathcal{Z}$; it is *G-invariant* when rho is the trivial representation, i.e. $\mathrm{phi}_{\mathrm{theta}}(g \cdot x) = \mathrm{phi}_{\mathrm{theta}}(x)$. In plagiarism detection the siamese head compares embeddings $z = \mathrm{phi}_{\mathrm{theta}}(x)$ and $z' = \mathrm{phi}_{\mathrm{theta}}(x')$, so invariance of $\mathrm{phi}_{\mathrm{theta}}$ directly implies that geometric variants of the same image collapse to the same embedding without any augmentation. Full theoretical treatment of these properties is deferred to Section 4.

Several recent lines of work target geometric robustness in vision transformers. Nordström et al. [5] introduce the Octic ViT, which enforces exact D_4-equivariance through group-structured

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Table 1: Comparison of approaches to geometric robustness in image matching.

Method	Group	Exactness	Strengths	Weaknesses
Dorin et al. [9]	None	Approximate	Flexible; strong features	No invariance guarantee
Octic ViT [5]	D_4	Exact	40% FLOPs reduction	Discrete; 90° only
Shift-Eq. ViT [6]	$\mathbb{Z}^2$	Exact	Exact shift invariance	No rotation equivariance
Harmformer [7]	$SE(2)$	Exact	Arbitrary angles + shifts	Higher compute cost

weight sharing and reduces FLOPs by 40% relative to a standard ViT of equal expressivity. Rojas-Gomez et al. [6] design a Shift-Equivariant ViT that achieves exact $\mathbb{Z}^2$-invariance via anti-aliased downsampling and circular padding, but does not address rotational equivariance. Karella et al. [7] propose Harmformer, which lifts transformers to the continuous group $SE(2)$ using harmonic feature maps and obtains equivariance to arbitrary in-plane rotations and translations at a higher compute cost. On the side of learned features and plagiarism-specific architectures, Oquab et al. [8] show that DINOv2 pre-training on large unlabeled image collections yields transferable representations, and Dorin et al. [9] establish a siamese ViT baseline for pairwise plagiarism detection with false positive rate (FPR) as the primary evaluation metric [10].

We propose a siamese plagiarism detector with G-equivariant ViT encoders that provides exact invariance under D_4, $\mathbb{Z}^2$, or $SE(2)$ without geometric augmentation. Relative to Dorin et al. [9], two elements are new: (i) an L^2 contrastive regularizer that penalizes the Euclidean distance between embeddings of genuine pairs and pushes impostor pairs beyond a margin, and (ii) a controlled augmentation ablation that pits geometric augmentation against equivariant inductive bias across three symmetry groups in one framework. Concretely, we train five encoder configurations - standard ViT with augmentation, DINOv2-pretrained ViT, Octic ViT (D_4), Shift-Equivariant ViT ($\mathbb{Z}^2$), and Harmformer ($SE(2)$) - initialized with DINOv2 weights on ImageNet-1K where architecturally compatible, each in two conditions: with and without the L^2 contrastive regularizer.

The experiment comprises 10 runs (5 encoders $\times$ {with, without} L^2 regularizer) evaluated on a class-balanced subset of DomainNet spanning six visual domains. Primary metrics are FPR and Recall at a fixed decision threshold; the secondary metric is F1. This design isolates the contribution of each inductive bias and tests whether exact geometric invariance reduces false positives relative to an augmentation-only baseline. Table 1 summarises the methods compared.

The remainder of this paper is organized as follows. Section 2 reviews related work on equivariant networks, siamese architectures, and plagiarism detection. Section 3 formalizes the detection task and evaluation protocol. Section 4 develops the theoretical background on group equivariance. Section 5 describes the siamese framework, encoder configurations, and the L^2 contrastive regularizer. Section 6 presents experimental results and ablations. Section 7 concludes with limitations and future directions.

2 Related Work

Siamese detectors for image matching and plagiarism. Bromley et al. [10] introduced siamese networks for signature verification: a shared-weight twin architecture with a learned distance metric that extends to any pairwise similarity task. Classical near-duplicate detection relied on compact local descriptors combined with geometric verification [11], an approach that breaks under large photometric or domain shifts. Dorin et al. [9] applied a siamese ViT with contrastive loss to image plagiarism detection on DomainNet, reporting FPR as the primary metric and using a cross-batch pairing strategy. We extend their protocol with equivariant encoders and an L^2 contrastive regularizer defined on Euclidean distances in the embedding space.

Equivariant and invariant architectures. Cohen and Welling [4] introduced G-CNNs that encode discrete symmetries into convolutional filters through group-structured weight sharing. Weiler and Cesa [12] generalised this idea to steerable E(2)-equivariant CNNs, covering continuous rotation and reflection groups within one framework. Bronstein et al. [3] treat equivariance as an inductive bias that improves sample efficiency and generalisation across domains. These architectures have been

evaluated mostly on classification and segmentation; to our knowledge, they have not been studied on pairwise image similarity in a siamese plagiarism-detection setting.

Equivariant Vision Transformers. Nordström et al. [5] replace standard ViT layers with D_4-equivariant variants, obtaining a 40% FLOPs reduction at accuracy comparable to the baseline ViT. Rojas-Gomez et al. [6] enforce exact shift-equivariance through adaptive polyphase sampling, introducing A-token, A-WSA, A-PMerge, and A-RPE modules without adding learnable parameters. Karella et al. [7] combine circular harmonic filters with transformer blocks to obtain continuous SE(2)-equivariance across arbitrary rotation angles. None of these architectures has been evaluated on pairwise image similarity or plagiarism detection, which leaves open whether geometric inductive bias reduces false positives on such tasks.

Self-supervised ViT pre-training and contrastive learning. DINOv2 [8] produces transferable visual features via self-supervised learning on large unlabeled image collections; we use it as a common pre-training recipe across all encoders so that performance differences reflect architectural choices rather than pre-training data. Chopra et al. [13] proposed contrastive loss for discriminative metric learning, minimising intra-class distance while maximising inter-class distance. Hadsell et al. [14] introduced the L^2 contrastive formulation we adopt: the loss depends only on Euclidean distances, so it is preserved under orthogonal transformations of the embedding space.

3 Problem Statement

1. General problem. Given a pair of images $(x_1, x_2) \in \mathcal{X} \times \mathcal{X}$, the task is binary classification: predict $y_i \in \{0, 1\}$ where $y_i = 1$ indicates that x_2 is a plagiarised derivative of x_1. Plagiarism is operationalised as the composition of a geometric transformation $g \in G$ with a photometric perturbation $\mathrm{pi} \in G_{\mathrm{photo}}$ (colour jitter, Gaussian blur, additive noise), yielding $x_2 = (g \circ \mathrm{pi})(x_1)$. The detector must correctly identify such pairs regardless of which transformation was applied.

2. Sample set. The labelled dataset is

$$\mathcal{D} = \big\{(x_{1,i},\, x_{2,i},\, y_i)\big\}_{i=1}^{N},$$

where $y_i \in \{0, 1\}$ and every image $x_i \in \mathcal{X} = [0, 1]^{3\times 224\times 224}$. Each sample is a triple: a query image, a candidate image, and a binary plagiarism label. The dataset is balanced across positive and negative pairs and stratified by visual domain.

3. Statistical hypothesis. Positive pairs are drawn from the joint distribution

$$p_+(x_1, x_2) = p(x_1)\, p_T(x_2 \mid x_1),$$

where $p_T(x_2 \mid x_1) = \int_{G\times G_{\mathrm{photo}}} \mathrm{delta}(x_2 - (g \circ \mathrm{pi})(x_1))\, d\mathrm{mu}(g, \mathrm{pi})$ is the mixture of transformation kernels over $G \cup G_{\mathrm{photo}}$. Negative pairs are obtained by sampling x_1 and x_2 independently from the marginal $p(x)$, so $p_-(x_1, x_2) = p(x_1)\, p(x_2)$. The detector must distinguish the two regimes on held-out data.

4. Measurement conditions. All experiments use the DomainNet dataset [15], which spans six visually distinct domains: real photograph, painting, clipart, quickdraw, infograph, and sketch. Images are resized to 224×224 pixels in 8-bit RGB and normalised per-channel using ImageNet mean and standard deviation. This multi-domain structure enables evaluation of generalisation across severe appearance shifts.

5. Sample restrictions. Positive pairs consist of a source image x and its transformed copy $(g \circ \mathrm{pi})(x)$, where $g \sim \mathrm{Uniform}(G)$ and $\mathrm{pi} \sim p_{\mathrm{photo}}$. Negative pairs are constructed from two independently sampled images drawn from the *same* DomainNet domain, making them hard negatives with shared visual statistics. The evaluation set contains 1200 pairs in total: 100 positives and 100 negatives for each of the six domains, ensuring balanced coverage.

6. Model. The full model is $f_{\mathrm{theta}} : \mathcal{X} \times \mathcal{X} \to [0, 1]$, decomposed as

$$f_{\mathrm{theta}}(x_1, x_2) = \mathrm{sigma}\big(h_{\mathrm{psi}}(\mathrm{phi}_{\mathrm{theta}}(x_1),\ \mathrm{phi}_{\mathrm{theta}}(x_2))\big), \tag{1}$$

where $\mathrm{phi}_{\mathrm{theta}} : \mathcal{X} \to \mathbb{R}^d$ is the encoder with shared weights, h_{psi} is the fusion module, and sigma denotes the sigmoid function. The siamese design ensures that both images are mapped to the same embedding space before comparison.

7. Class restrictions. The encoder $\mathrm{phi}_{\mathrm{theta}}$ must be G-equivariant with a G-invariant pooling step so that the resulting embedding satisfies $\mathrm{phi}_{\mathrm{theta}}(g \cdot x) = \mathrm{phi}_{\mathrm{theta}}(x)$ for all $g \in G$. This invariance guarantees that geometric variants of the same image collapse to identical embeddings without data augmentation. Additionally, the fusion module h_{psi} must be symmetric: $h_{\mathrm{psi}}(z_1, z_2) = h_{\mathrm{psi}}(z_2, z_1)$, ensuring that the plagiarism score does not depend on the ordering of the two inputs.

8. Loss / criterion. Training minimises the composite objective

$$\mathcal{L} = \mathcal{L}_{\mathrm{BCE}} + \mathrm{lambda}\, \mathcal{L}^{L^2}_{\mathrm{contr}}, \tag{2}$$

where $\mathcal{L}_{\mathrm{BCE}}$ is weighted binary cross-entropy with $w_+ = 0.3$ and $w_- = 0.7$, and the contrastive regulariser is

$$\mathcal{L}^{L^2}_{\mathrm{contr}}(z_1, z_2, y_{ij}) = y_{ij}\, \|z_1 - z_2\|_2^2 + (1 - y_{ij})\, \max\big(0,\ m - \|z_1 - z_2\|_2\big)^2. \tag{3}$$

Setting $\mathrm{lambda} = 0$ recovers the variant trained without contrastive regularisation, which serves as a baseline to isolate the contribution of the geometric loss term.

9. Cross-validation procedure. Generalisation is estimated via stratified five-fold cross-validation over the DomainNet evaluation pool, with stratification by domain to ensure that every fold contains examples from all six domains. Hyper-parameters are selected on the validation fold, and test metrics are averaged across folds. This procedure is applied identically to all encoder configurations to ensure comparability.

10. Solution restrictions. All experiments are conducted on a single GPU with at most 48 GB of VRAM, and fine-tuning is limited to at most 30 epochs. The encoder backbone $\mathrm{phi}_{\mathrm{theta}}$ is frozen throughout training; only the fusion head h_{psi} is updated. These constraints reflect a realistic resource budget and ensure that differences in performance reflect architectural choices rather than training cost.

11. External quality criteria. The primary metrics are **FPR** (false positive rate at the threshold that maximises F1 on the validation fold) and **Recall** reported per transformation class: colour jitter (CJ), Gaussian noise (GN), Gaussian blur (GS), rotation by 90° (R90), 180° (R180), 270° (R270), watermark (WM), and combinations thereof. Secondary metrics are F1 score and t-SNE separability of embeddings z on DomainNet. Stratified recall directly measures whether a given encoder is robust to specific transformation types.

12. Argmin. The optimal encoder and fusion parameters are given by

$$\hat{\mathrm{theta}} = \arg\min_{\mathrm{theta}}\ \mathbb{E}_{(x_1,\, x_2,\, y_i)\, \sim\, \mathcal{D}}\, \mathcal{L}(\mathrm{theta}). \tag{4}$$

In practice, this expectation is approximated by the empirical mean over mini-batches, and $\hat{\mathrm{theta}}$ is taken as the checkpoint with the lowest validation loss across the five cross-validation folds.

4 Preliminaries: Equivariance

4.1 Group Actions and Representations

Definition 1 (Group action)**.** *A* left action *of a group G on a set $\mathcal{X}$ is a map $G \times \mathcal{X} \to \mathcal{X}$, $(g, x) \mapsto g \cdot x$, satisfying $e \cdot x = x$ and $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$, $x \in \mathcal{X}$.*

Definition 2 (Group representation)**.** *A* (linear) representation *of G on a vector space V is a group homomorphism* $\mathrm{rho}\colon G \to \mathrm{GL}(V)$.

A representation makes concrete how G acts on a feature space: each group element g is assigned an invertible linear map $\mathrm{rho}(g)$ on V, and group multiplication is preserved by $\mathrm{rho}(gh) = \mathrm{rho}(g)\mathrm{rho}(h)$.

4.2 Equivariance and Invariance

Definition 3 (G-Equivariance)**.** *A map* $\mathrm{phi}\colon \mathcal{X} \to \mathcal{Y}$ *is* G-equivariant *with respect to representations* $\mathrm{rho}_{\mathcal{X}}$ *on* $\mathcal{X}$ *and* $\mathrm{rho}_{\mathcal{Y}}$ *on* $\mathcal{Y}$ *if*

$$\mathrm{phi}\big(\mathrm{rho}_{\mathcal{X}}(g) \cdot x\big) = \mathrm{rho}_{\mathcal{Y}}(g) \cdot \mathrm{phi}(x) \quad \forall\, g \in G,\ x \in \mathcal{X}.$$

Definition 4 (G-Invariance)**.** phi *is* G-invariant *if* $\mathrm{rho}_{\mathcal{Y}}$ *is the trivial representation:* $\mathrm{phi}(\mathrm{rho}_{\mathcal{X}}(g) \cdot x) = \mathrm{phi}(x)$ *for all* $g \in G$.

Invariance is thus a special case of equivariance in which the output space carries only the trivial action of G. For plagiarism detection, a G-invariant encoder $\mathrm{phi}_{\mathrm{theta}}$ satisfies $\mathrm{phi}_{\mathrm{theta}}(g \cdot x) = \mathrm{phi}_{\mathrm{theta}}(x)$ for every $g \in G$, so embeddings are identical regardless of which group element was applied to the source image. This removes the need for group-specific augmentation at training time and yields an exact invariance guarantee at inference time.

4.3 Irreducible Representations and Schur's Lemma

Lemma 1 (Schur)**.** *Let* $\mathrm{rho}_1\colon G \to \mathrm{GL}(V_1)$ *and* $\mathrm{rho}_2\colon G \to \mathrm{GL}(V_2)$ *be irreducible representations of a group* G*. Any* G*-equivariant linear map* $\mathrm{phi}\colon V_1 \to V_2$ *is either the zero map or an isomorphism.*

A consequence of Schur's lemma is that G-equivariant linear maps between fully decomposed representations are block-diagonal with respect to the irreducible decomposition: each block acts within one irreducible component and cannot mix distinct irreducible types. This constraint shrinks the parameter space of equivariant layers relative to unconstrained linear maps, and it motivates the architecture of Octic ViT [5], where each transformer layer operates independently on each irreducible component of D_4 so the block-diagonal structure holds by construction.

4.4 Symmetry Groups in This Work

The dihedral group D_4. $D_4 = \langle r, s \mid r^4 = s^2 = e,\ srs = r^{-1} \rangle$ is the symmetry group of the square, of order 8. It acts on images by rotating and reflecting the spatial axes by multiples of 90°. D_4 has five irreducible representations over $\mathbb{R}$: four one-dimensional (A_1, A_2, B_1, B_2) and one two-dimensional (E); any D_4-representation decomposes as $V \cong \bigoplus_i n_i V_i$ over these five types.

The discrete translation group $\mathbb{Z}^2$. The group $(\mathbb{Z}^2, +)$ acts on an $H \times W$ image by cyclic pixel shifts: $(t_{a,b} \cdot x)[i, j, c] = x[(i + a) \bmod H,\ (j + b) \bmod W,\ c]$. This action is relevant when plagiarists crop and re-centre images. In contrast to D_4, the group $\mathbb{Z}^2$ is infinite (or large-but-finite on discrete images), and exact equivariance is harder to enforce in the ViT patch-based setting.

The special Euclidean group $SE(2)$. $SE(2) = SO(2) \ltimes \mathbb{R}^2$ is the group of rigid motions of the plane: an element (R_{alpha}, t) rotates spatial coordinates by angle alpha and translates by $t \in \mathbb{R}^2$. Because D_4 covers only multiples of 90° while $SE(2)$ admits arbitrary rotation angles, $SE(2)$-equivariance is more general but also harder to realise exactly on a discrete pixel grid.

5 Method

5.1 Siamese Framework

The detector follows the siamese architecture of Bromley et al. [10]. Two branches share one encoder $\mathrm{phi}_{\mathrm{theta}}$ and process the query image x_1 and the candidate image x_2 in parallel to produce embeddings $z_1 = \mathrm{phi}_{\mathrm{theta}}(x_1)$ and $z_2 = \mathrm{phi}_{\mathrm{theta}}(x_2)$. A fusion module h_{psi} combines them into a scalar score, and a sigmoid maps the score to a plagiarism probability. The full model (already introduced in §3) is

$$f_{\mathrm{theta}}(x_1, x_2) = \mathrm{sigma}\big(h_{\mathrm{psi}}(\mathrm{phi}_{\mathrm{theta}}(x_1),\ \mathrm{phi}_{\mathrm{theta}}(x_2))\big). \tag{5}$$

Weight sharing maps x_1 and x_2 into the same feature space; without it the two branches would produce vectors in unrelated coordinate systems and any pairwise comparison would be ill-defined.

5.2 Encoders

ViT-L/16 (augmented). The baseline encoder is a standard ViT-L/16 [1]. Each 224×224 input is split into $14 \times 14 = 196$ non-overlapping 16×16 patches; each patch is linearly projected to $\mathbb{R}^{1024}$, and the resulting sequence, together with a prepended `[CLS]` token, is passed through 24 transformer blocks with multi-head self-attention. The `[CLS]` token after the final layer norm is taken as the image embedding $z \in \mathbb{R}^{1024}$. This encoder has no equivariance guarantee. At training time it receives both photometric augmentations (block P) and geometric augmentations (block G); see §5.5.

ViT-L/16 (baseline). Architecturally identical to ViT-L/16 (augmented), but trained with block P only; block G geometric transforms are withheld. This is the key ablation: if the augmentation-free variant matches the equivariant encoders, then geometric augmentation alone suffices and architectural inductive bias adds nothing beyond what augmentation already provides.

Octic ViT (D_4-equivariant). Nordström et al. [5] replace each ViT layer with a D_4-equivariant variant. By Schur's lemma (§4), equivariant linear maps act independently on each irreducible component, so the embedding space decomposes as

$$\mathbb{R}^d \;\cong\; \underbrace{(\mathbb{R}^{d/8})_{A_1} \oplus (\mathbb{R}^{d/8})_{A_2} \oplus (\mathbb{R}^{d/8})_{B_1} \oplus (\mathbb{R}^{d/8})_{B_2}}_{\text{four 1-dim irreps}} \oplus \underbrace{(\mathbb{R}^{d/4})_E}_{\text{one 2-dim irrep}} . \tag{6}$$

Invariant pooling projects onto the trivial component A_1 to give a D_4-invariant output z. Trained with block P only.

Shift-Equivariant ViT ($\mathbb{Z}^2$-equivariant). Rojas-Gomez et al. [6] enforce exact cyclic shift-equivariance by replacing four ViT modules with adaptive polyphase variants: an A-token module (per-sample polyphase offset selection), A-WSA (adaptive window self-attention partition), A-PMerge (polyphase downsampling), and A-RPE (circular relative position encoding). The four modifications yield exact shift-equivariance without introducing extra learnable parameters. Trained with block P only.

Harmformer ($SE(2)$-equivariant). Karella et al. [7] combine circular harmonic filters (steerable feature maps) with transformer self-attention blocks to obtain equivariance under continuous roto-translations. Compared with Octic ViT, which is restricted to 90° multiples of the dihedral group, Harmformer covers arbitrary rotation angles; the stronger guarantee comes at a higher computational cost. Trained with block P only.

All encoders are pre-trained on ImageNet-1K using the DINOv2 [8] self-supervised recipe.

5.3 Fusion Module

The fusion module combines two embeddings $z_1, z_2 \in \mathbb{R}^d$ into a scalar via a two-layer MLP applied to element-wise features:

$$h_{\text{psi}}(z_1, z_2) = W_2\, \mathrm{ReLU}\big(W_1\, [|z_1 - z_2| \parallel z_1 \odot z_2] + b_1\big) + b_2, \tag{7}$$

where $|\cdot|$ denotes element-wise absolute value, $\odot$ element-wise product, and $[\cdot\|\cdot]$ concatenation.

Proposition 1 (Symmetry of the fusion module)**.** $h_{\text{psi}}(z_1, z_2) = h_{\text{psi}}(z_2, z_1)$.

Proof. $|z_1 - z_2| = |z_2 - z_1|$ and $z_1 \odot z_2 = z_2 \odot z_1$; therefore the concatenated input to the MLP is the same under transposition. □

5.4 L^2 Contrastive Regularizer

The contrastive regularizer (introduced in §3) is

$$\mathcal{L}^{L^2}_{\text{contr}}(z_1, z_2, y_{ij}) = y_{ij}\, \|z_1 - z_2\|_2^2 + (1 - y_{ij})\, \max\big(0,\, m - \|z_1 - z_2\|_2\big)^2. \tag{8}$$

We use L^2 distance rather than cosine similarity because it is better aligned with the group-invariant geometry of the embedding space. Following Hadsell et al. [14], the quantity $\|z_1 - z_2\|_2$ is an

isometric invariant: it is preserved by any orthogonal transformation of $\mathbb{R}^d$. This matches G-invariant encoders that may differ only by an isometry of the embedding space. Cosine similarity, in contrast, depends on vector norms, so it is not preserved by general orthogonal maps.

Proposition 2 (Isometry invariance of $\mathcal{L}^{L^2}_{\text{contr}}$)**.** *Let $U\colon \mathbb{R}^d \to \mathbb{R}^d$ be an orthogonal map. Then $\mathcal{L}^{L^2}_{\text{contr}}(Uz_1, Uz_2, y_{ij}) = \mathcal{L}^{L^2}_{\text{contr}}(z_1, z_2, y_{ij})$.*

Proof. $\|Uz_1 - Uz_2\|_2 = \|U(z_1 - z_2)\|_2 = \|z_1 - z_2\|_2$, since orthogonal maps preserve the L^2 norm. □

For each encoder we compare two conditions: $\text{lambda} > 0$ (with L^2 regularizer) and $\text{lambda} = 0$ (without).

5.5 Augmentation Blocks

Block P (photometric augmentations) is applied to all encoders during training:

Transform	Parameters
ColorJitter	brightness=0.2, contrast=0.2, saturation=0.1, hue=0.05
RandomGrayscale	$p = 0.05$
GaussianBlur	kernel=3, $p = 0.1$
ImageNet normalization	$\text{mu} = (0.485, 0.456, 0.406)$, $\text{sigma} = (0.229, 0.224, 0.225)$

Block G (geometric augmentations) is applied only to ViT-L/16 (augmented):

Transform	Parameters
RandomHorizontalFlip	$p = 0.5$
RandomVerticalFlip	$p = 0.5$
RandomChoice of rotation	$\{0°, 90°, 180°, 270°\}$
RandomResizedCrop	scale=$(0.7, 1.0)$, ratio=$(0.9, 1.1)$

Equivariant encoders by design produce the same embedding for any $g \in G$ applied to the input, so block G augmentation on those encoders generates pairs with zero gradient for the head: harmless but redundant. The experimental hypothesis is therefore that block G can be dropped for equivariant encoders without degrading detection performance.

5.6 Theoretical Properties: Invariance Propagation

Proposition 3 (Invariance propagation)**.** *If $\text{phi}_{\text{theta}}$ is G-invariant, then $f_{\text{theta}}(g \cdot x_1, g \cdot x_2) = f_{\text{theta}}(x_1, x_2)$ for all $g \in G$ and all $x_1, x_2 \in \mathcal{X}$.*

Proof. G-invariance of $\text{phi}_{\text{theta}}$ gives $\text{phi}_{\text{theta}}(g \cdot x_i) = \text{phi}_{\text{theta}}(x_i)$ for $i = 1, 2$. Since h_{psi} and sigma are deterministic functions of their inputs, the output of f_{theta} is unchanged. □

The plagiarism score is thus the same for every $g \in G$ applied jointly to both inputs, which is the invariance required for geometric plagiarism detection.

6 Experiments

6.1 Dataset and Pre-training

All encoders are pre-trained on COCO2017 with the DINOv2 [8] self-supervised recipe. Pre-training is done separately per encoder architecture when DINOv2-compatible weights exist; for architectures without publicly available DINOv2 checkpoints, we initialize from a ViT-L/16 DINOv2 checkpoint and adapt the weights.

Evaluation uses the DomainNet benchmark [15], which contains images from six visually diverse domains: real photograph, painting, clipart, quickdraw, infograph, and sketch. We build a balanced test set of 1200 pairs: 100 positive and 100 negative pairs per domain (600 positives and 600 negatives in total). Each image is resized to 224×224 pixels, giving $3 \times 224 \times 224 = 150{,}528$ scalar features per image. A positive pair is a source image together with a transformed copy; a negative pair is two distinct images drawn from the same domain (hard negatives).

6.2 Training Pipeline

The experiment consists of 10 training runs: 5 encoders $\times$ $\{\mathrm{lambda} = 0$ (without L^2 regulariser), $\mathrm{lambda} > 0$ (with L^2 regulariser)$\}$.

In each run we load base images from the training split and build pairs via the cross-batch strategy of Dorin et al. [9]: a batch of B images is augmented to form two views per image (block P for all encoders, and additionally block G for ViT-L/16 augmented), and all B^2 pairwise combinations are formed with labels $y_{ij} = \mathbb{1}[i = j]$. Embeddings are computed through the frozen encoder and the fusion module, the composite loss $\mathcal{L}$ is evaluated, and AdamW updates the fusion parameters psi. Hyperparameters are listed in Table 2.

Table 2: Training hyperparameters.

Parameter	Value
Optimizer	AdamW
Learning rate	10^{-3}
Weight decay	10^{-2}
LR scheduler	MultiStepLR, milestones $[5, 10]$, $\mathrm{gamma} = 0.5$
BCE weights	$w_+ = 0.3$, $w_- = 0.7$
Epochs	15
Contrastive margin m	1.0
Contrastive weight lambda (when enabled)	0.1
Batch size B	32

6.3 Evaluation Protocol

For each (encoder, $\pm L^2$) run, we train on the training split, pick a decision threshold on the validation fold by maximising F1, and then compute final metrics on the held-out test set.

Primary metrics are false positive rate (FPR) and Recall, both stratified by transformation class: ColorJitter (CJ), Gaussian Noise (GN), Grayscale (GS), R90, R180, R270, Watermark (WM), and their pairwise combinations. FPR is reported for comparability with Dorin et al. [9]. Secondary metrics include the F1 score and a t-SNE visualisation of encoder embeddings on the full DomainNet evaluation set, with the six domains shown as distinct point colours.

- **Figure 1.** Siamese architecture schema: encoder branches, fusion module, sigmoid head. (Axes: not applicable.)
- **Figure 2.** Diagrams of D_4, $\mathbb{Z}^2$, $SE(2)$ with example actions on a sample image. (Axes: spatial coordinates of the image under each transformation.)
- **Figure 3** (source data plot). Distribution of DomainNet evaluation pairs. x-axis: domain name; y-axis: pair count.
- **Figure 4** (main message). FPR by encoder and regularisation condition. x-axis: encoder name; y-axis: FPR; grouped bars: $\mathrm{lambda} = 0$ vs. $\mathrm{lambda} > 0$.
- **Figure 5** (main message). Recall stratified by transformation class. x-axis: transformation class (CJ, GN, GS, R90, R180, R270, WM, combos); y-axis: Recall; bars grouped by encoder.
- **Figure 6.** t-SNE of encoder embeddings on DomainNet. Five subplots (one per encoder); point colours: six DomainNet domains.

In addition to Table 1 (§ 1) and Table 2 (§ 6):

- **Table 3.** Main results: FPR for each (encoder, regularisation) run.
- **Table 4.** Recall stratified by transformation class for each (encoder, regularisation) run.

6.4 Results

7 Conclusion

We presented a siamese plagiarism detector that benchmarks five encoder configurations spanning three symmetry groups, D_4 (Octic ViT), $\mathbb{Z}^2$ (Shift-Equivariant ViT), and $SE(2)$ (Harmformer), against a plain ViT-L/16 baseline. Fixing the DINOv2 pre-training protocol and varying only the geometric inductive bias and the presence of an L^2 contrastive regularizer lets the framework separate the contribution of built-in G-invariance from that of data augmentation across 10 runs. Section 5 establishes three properties of the setup: symmetry of the fusion module, isometry invariance of the L^2 contrastive loss, and invariance propagation through the full siamese pipeline. Whether G-invariant encoders trained without block G augmentations match or exceed a ViT-L/16 trained with full augmentation is the empirical question left to the DomainNet evaluation.

The experimental runs (§6.4) will be completed in subproject C: implementing Shift-Equivariant ViT and Harmformer, pre-training all five encoders on ImageNet-1K, and executing the 10 siamese fine-tuning experiments. We will report FPR and Recall stratified by transformation class on the balanced DomainNet test set. We expect the equivariant encoders to support the inductive bias hypothesis: reduced reliance on geometric augmentation with detection performance preserved.

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